CCS3002 Numerical Methods
12th Project Requirements

Project Requirements

1. Teams of 2 students.
2. Implement using any programming language.
3. Choose from each chapter one method, except for chapter 2 implement both methods for *bonus*.
   ***For example**, you can submit* (**CH1**: Newton Raphson, **CH2:** Jacobi and Gauss, **CH3:** Newton Forward/Backward Divided Difference, **CH4:** Midpoint (Basic + Composite).
4. Make the inputs as dynamic as possible. ***For example:*** I can enter any equation or any number of equations with the initial point(s).
5. Do the checks, ***for example***, in bisection make sure that the f(a) and f(b) are opposite signs.

Chapter 1
- Bisection
- Secant
- Newton – Raphson
- Simple Iteration

Chapter 2
- Jacobi
- Gauss-Seidel

Chapter 3
- Lagrange's Interpolation
- Newton Forward/Backward Difference (NF/BD)
- Newton Forward/Backward Divided Difference (NF/BDD)

Chapter 4
- Midpoint (Basic + Composite)
- Trapezoidal (Basic + Composite)
- Simpson's (Basic + Composite)
- Gaussian Quadrature
- Numerical Differentiation

Grading

The 10 marks evaluation will be applied to the following criteria:

a. Your ability to successfully run the code.
b. Dynamic Inputs.
c. Correct output.
d. Error handling, checks and special conditions.
e. Discussion to ensure you fully understand the code.

Deadline **Discussion will be on week 12 on section time.**

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